

# USHNA

Unified Steam, Hydraulics & Nodal Analysis

## A Physics-First Digital Twin for Well-to-Surface Optimization of CSS and SRP Operations

Baghewala Heavy Oil Field, Rajasthan

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*Every setpoint this system emits is traceable to a governing equation and a named physical parameter. Machine learning estimates parameters, corrects known model gaps, and accelerates solvers. It never replaces the physics, and it never issues a recommendation that an engineer cannot verify on paper.*

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|--------------------------|-------------------|
| <b>Problem Statement</b> | 26120             |
| <b>Organization</b>      | Oil India Limited |
| <b>Category</b>          | Software          |
| <b>Theme</b>             | Smart Automation  |

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## 1 Problem Understanding

Baghewala produces heavy crude of 17–19° API from the Jodhpur Sandstone. The reservoir is cold (46–48°C), under-pressured, asphaltene-rich, and the oil has very poor mobility under primary recovery. Two interventions keep the field alive: Cyclic Steam Stimulation to reduce viscosity, and Sucker Rod Pumps to lift the fluid. Both are indispensable.

### 1.1 The actual failure mode

These two systems are physically coupled but operationally divorced.

A CSS cycle heats the near-wellbore region; viscosity collapses; the pump runs easily and production spikes. Over the following weeks the heated zone cools by conduction to the over- and underburden and by convection with produced fluid. Viscosity climbs back. The SRP settings that were correct on day 5 are actively destructive by day 40 — but nobody re-tunes them, because tuning is manual and reactive.

**Rod floating is the signature consequence, and it is a viscosity problem, not a pump problem.** On the downstroke the rod string must displace fluid. When viscous drag exceeds the buoyed weight of the rods, the string stops falling while the surface unit continues its descent. The rods go into compression, buckle against the tubing, and then slam when they re-engage — impact loading. Every rod failure, every pump unsettling, and a large share of maintenance cost at Baghewala is downstream of one uncorrected event: *viscosity rose and pumping speed did not come down*.

On the CSS side the cost metric is Steam–Oil Ratio. Excess steam volume or an over-long soak burns fuel for barrels that would have arrived anyway; too little and the well cools before it has paid back the injection. The production cut-off decision — when to stop producing and re-inject — carries the largest economic leverage in the entire operation and is currently made by habit.

### 1.2 Three coupled timescales

| Layer         | Timescale         | Decisions owned                                      |
|---------------|-------------------|------------------------------------------------------|
| Reservoir     | days – weeks      | steam volume, injection pressure, soak time, cut-off |
| Wellbore      | hours             | fluid temperature, viscosity at pump, inflow, gas    |
| Surface (SRP) | seconds – minutes | stroke length, SPM, VFD velocity profile             |

The wellbore layer is the bridge. It converts reservoir thermal state into a *viscosity-at-the-pump* number, which is precisely the quantity the surface controller needs and does not currently have.

**The coupling variable is  $\mu(T_{\text{pump}})$ .** A solution that builds a reservoir dashboard and a pump dashboard side by side has not integrated anything. Integration means one scalar flowing from the thermal model into the control law, continuously.

## 2 Design Philosophy: Why Not a Black Box

The temptation with a data-rich problem statement is to pour history into a gradient-boosted model and present feature importances. For this field that approach fails on four counts.

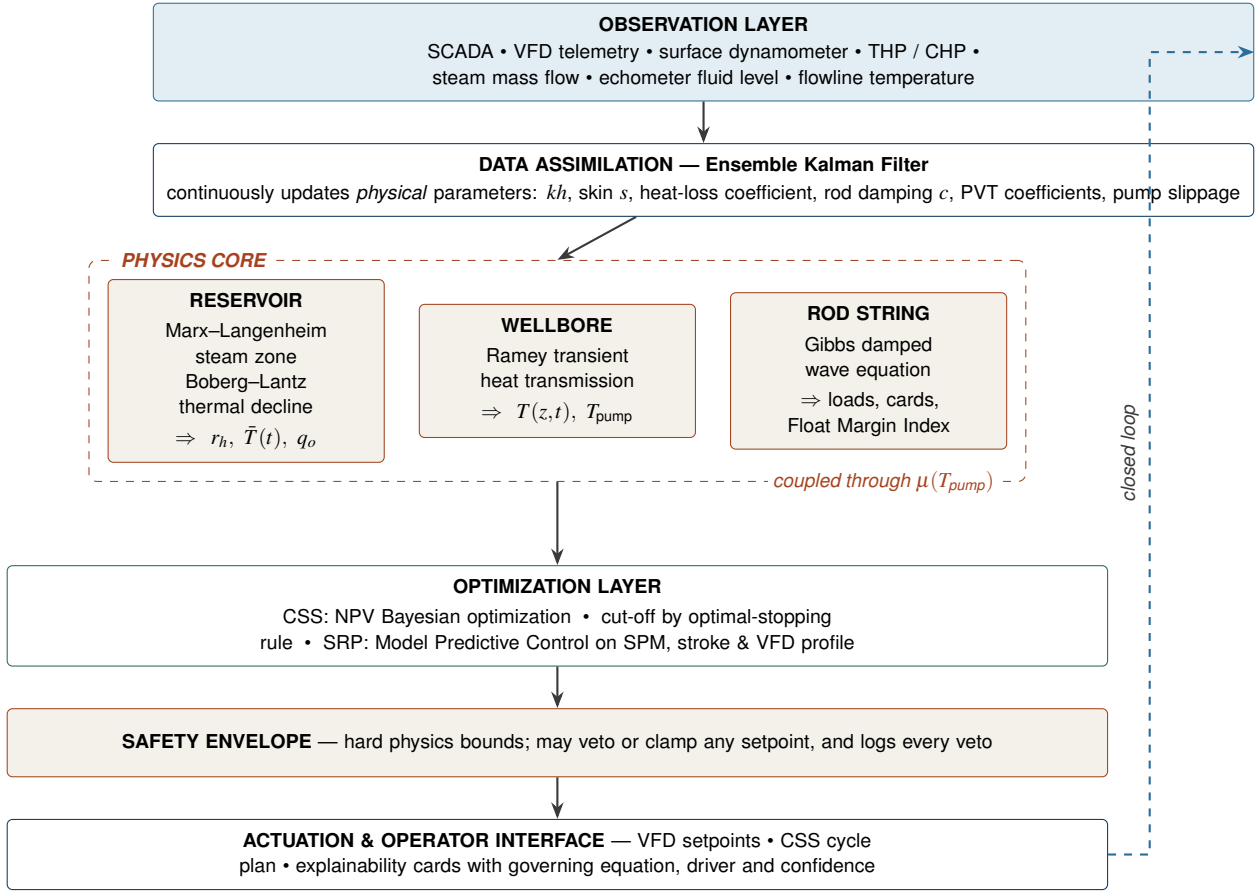
- 1. The data is sparse where it matters.** A CSS field accumulates perhaps 30–80 completed cycles of history. That is a small-sample regime. A deep model will memorise it; a physics model with a handful of estimated parameters will generalise from it.
- 2. The task requires extrapolation by definition.** Optimising CSS means proposing steam volumes and soak times *outside historical practice* — that is the entire point. A statistical model interpolates its training

envelope and guesses beyond it. A conservation equation computes the answer wherever you evaluate it.

3. **It will not be trusted, and therefore not deployed.** Oil India will not permit an unexplainable model to command a rod string. A recommendation that cannot be audited by the field engineer on duty is a recommendation that gets overridden every time.
4. **It fails the domain expert's question.** The evaluators are petroleum engineers. "What damping coefficient did you use and how did you obtain it?" must have an answer containing a number and a method.

**The contract adopted throughout this design.** Physics owns the state equations. Learning is confined to four roles: *(i)* estimating physical parameters from live data, *(ii)* learning bounded corrections to known model gaps, *(iii)* building fast differentiable surrogates of validated solvers, and *(iv)* discovering closed-form correlations. Every one of those outputs is a named quantity with units and an uncertainty band — not a weight in a hidden layer.

### 3 System Architecture



The loop closes: actuation changes well behaviour, observation captures it, assimilation corrects the parameters, and the physics core improves. A model that does not self-correct against its own asset is a simulator, not a twin.

### 4 The Physics Core

#### 4.1 Reservoir: CSS thermal behaviour and productivity

##### 4.1.1 Injection phase — Marx–Langenheim

The energy balance on the growing steam zone, with conductive losses to over- and underburden, gives heated volume as a function of injected steam:

$$V_s(t) = \frac{Q_i M_R \Delta T}{4 k_{ob} \rho_{ob} c_{ob} (\Delta T)^2} F(t_D), \quad t_D = \frac{4 k_{ob} t}{M_R h^2 \alpha_{ob}} \quad (1)$$

where  $F(t_D)$  carries the standard  $e^{t_D} \text{erfc} \sqrt{t_D}$  loss term. The model returns heated-zone radius  $r_h$  and steam temperature  $T_s$ , the latter read from steam tables at the injection pressure.

##### 4.1.2 Soak and production — Boberg–Lantz

Average heated-zone temperature decays as

$$\bar{T}(t) = T_R + (T_s - T_R) \left[ f_{VD} f_{HD} (1 - \delta) \right] \quad (2)$$

with  $f_{VD}$  and  $f_{HD}$  the vertical and radial conduction unit solutions, and  $\delta$  accounting for energy carried away with produced fluid.

That  $\delta$  term is the whole economics of CSS in one coefficient: **producing faster cools the heated zone faster**. It is the formal reason cut-off timing and SRP pumping speed cannot be optimised independently — the surface controller is directly spending the thermal asset that the injection paid for.

#### 4.1.3 Inflow — radial composite

A hot inner annulus surrounded by cold reservoir:

$$q_o = \frac{2\pi k k_{ro} h (\bar{P}_R - P_{wf})}{\mu_h \ln \frac{r_h}{r_w} + \mu_c \ln \frac{r_e}{r_h} + s} \quad (3)$$

Because  $\mu_h \ll \mu_c$ , this reproduces the observed post-steam production spike and its decay without any curve fitting. A Vogel correction handles two-phase inflow below bubble point.

**This entire chain evaluates in microseconds.** That is what makes thousands of optimizer iterations feasible on field hardware. A full commercial thermal simulator is accurate but cannot sit inside a real-time control loop; this formulation can, and it is validated *against* such a simulator rather than replacing it blindly.

## 4.2 Viscosity: the coupling law

ASTM D341 / Walther double-logarithmic form, fitted to Baghewala PVT:

$$\log_{10} \log_{10} (v + 0.7) = A - B \log_{10} T \quad (4)$$

For 17–19° API crude, kinematic viscosity swings by roughly two orders of magnitude between  $T_s$  and  $T_R$ . An asphaltene term is carried alongside: below an onset temperature  $T_{\text{onset}}$  the fluid becomes measurably shear-thinning and deposition risk rises, so a power-law index  $n$  is tracked and the regime  $T < T_{\text{onset}}$  is flagged as a distinct operating state with its own constraint set.

## 4.3 Wellbore: Ramey transient heat transmission

Ramey's solution gives tubing fluid temperature  $T(z, t)$  as a function of flow rate, time since start of production, and formation thermal conductivity. Evaluated at pump setting depth it yields  $T_{\text{pump}}$ , and hence  $\mu_{\text{pump}}$  through the Walther fit.

This is the bridge between reservoir and surface. It is also the reason a rate change propagates into rod loading with a lag of *hours* — a lag that any naive reactive controller will miss, causing it to oscillate. The predictive controller in Section 6 handles it explicitly because the lag is in the model.

## 4.4 Rod string: the Gibbs damped wave equation

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} - c \frac{\partial u}{\partial t}, \quad a = \sqrt{\frac{Eg}{\rho}} \approx 4900 \text{ m/s} \quad (5)$$

It is used in *both* directions, and that is the operational trick:

- **Inverse solve.** Surface dynamometer card → downhole pump card, via the everted solution. This converts a cheap, already-installed surface sensor into a downhole measurement, on every single stroke, with no additional hardware and no well intervention.

- **Forward solve.** Given  $\mu_{\text{pump}}$ , SPM, stroke length and rod taper, predict loads and the resulting card — which is what makes what-if analysis and predictive control possible.

#### 4.5 Rod float: an explicit design criterion

This is the sharpest result in the design, and it is worth stating completely.

Viscous drag on a rod moving axially within a concentric annulus of fluid, in the laminar regime, per unit length:

$$f_{\text{drag}} \approx \frac{2\pi\mu v_{\text{rod}}}{\ln(D_t/D_r)} \quad (6)$$

On the downstroke, float initiates when accumulated drag plus mechanical friction exceeds the buoyed weight of the rod string above that depth. Define a continuous, monitorable index:

$$\text{FMI}(z) = \frac{W_{\text{buoyant}}(z) - F_{\text{drag}}(z) - F_{\text{fric}}(z)}{W_{\text{buoyant}}(z)}$$

**Float Margin Index.** Rod float onset corresponds to  $\text{FMI}(z) \leq 0$  anywhere in the string. The operating limit is set at  $\text{FMI} > 0.15$  to retain margin.

Substituting  $v_{\text{rod}} \propto S \cdot N$  (stroke length  $\times$  strokes per minute) into the drag term gives the governing design relation:

$$(S \cdot N)_{\text{max}} \propto \frac{1}{\mu(T_{\text{pump}})}$$

**Maximum safe pumping speed is inversely proportional to viscosity at the pump.** As the reservoir cools through the CSS cycle, physics dictates exactly how far to back off SPM. No training data is required, no black box is involved, and a field engineer can verify the number on paper in two minutes.

This converts “rod floating is a recurring mystery failure” into a computed scalar with a setpoint attached to it. Two further constraints are layered on top:

- **Lubinski buckling** analysis for the segment in compression, giving the onset of helical buckling and the resulting tubing contact loads.
- **Modified Goodman** fatigue envelope on the tension side, converting the computed load range into consumed rod life — which is what turns “reduced rod failures” from an aspiration into a tracked number.

## 5 The Learning Layer

Four tiers, ordered from most to least constrained. Every tier produces interpretable output.

### 5.1 Tier 1 — Data assimilation by Ensemble Kalman Filter

This is the component that makes the system a *twin* rather than a simulator. An ensemble of physics models is maintained, and live measurements continuously nudge the *uncertain physical parameters* toward agreement with the real well.

| Estimated parameter              | Units              | Updated from                       |
|----------------------------------|--------------------|------------------------------------|
| Permeability-thickness $kh$      | mD·m               | rate and pressure history          |
| Skin $s$                         | —                  | rate and pressure history          |
| Overburden heat loss coefficient | W/m <sup>2</sup> K | observed temperature decline       |
| Rod damping coefficient $c$      | 1/s                | dynamometer card fit               |
| Viscosity coefficients $A, B$    | —                  | card, rate and temperature jointly |
| Pump slippage / leakage          | fraction           | fillage versus measured rate       |

**The learning is itself a diagnostic.** Every estimated quantity has a name, a unit, a physically plausible prior and a posterior uncertainty band. When skin drifts steadily upward cycle over cycle, that is not a model artefact — that is asphaltene deposition near the wellbore, and the system reports it as such, with a confidence interval. This is the direct answer to “is this a black box?”: a live table of named physical parameters with error bars.

### 5.2 Tier 2 — Bounded residual learning with Gaussian Processes

Following the Kennedy–O’Hagan calibration framework:

$$y_{\text{obs}} = f_{\text{physics}}(x, \theta) + \delta(x) + \varepsilon \quad (7)$$

A Gaussian Process learns *only the discrepancy*  $\delta(x)$ , hard-bounded to  $\pm 15\%$  of the physics prediction. Three consequences follow:

- Physics remains the backbone. The ML component is structurally incapable of producing a physically absurd answer, because it can only perturb a valid one by a bounded amount.
- Gaussian Processes give calibrated uncertainty natively, so every prediction ships with a confidence interval rather than a bare point estimate.
- Plotting *where*  $\delta$  is large diagnoses which physical assumption is failing. A large residual appearing only at high steam volumes, for instance, points to thief-zone override that the one-dimensional thermal model does not capture — an actionable finding about the reservoir, not a modelling nuisance.

### 5.3 Tier 3 — Physics-Informed Neural Network as a fast surrogate

There is exactly one place a neural network earns its keep here: the two-dimensional axisymmetric reservoir thermal field, where a full numerical solve is too slow for an optimizer performing  $10^4$  evaluations. The PINN is trained with the heat equation residual inside the loss function:

$$\mathcal{L} = \underbrace{\|\rho c \partial_t T - \nabla \cdot (k \nabla T) + q\|^2}_{\text{PDE residual}} + \mathcal{L}_{BC} + \lambda \mathcal{L}_{\text{data}} \quad (8)$$

**Its role is stated honestly.** The numerical solver remains ground truth; the PINN is a compressed, differentiable copy of it, validated against it. A candidate network that violates the energy balance is rejected at validation and never reaches deployment. The real prize is differentiability — it supplies exact analytic gradients to the optimizer, which is what makes gradient-based CSS design tractable at all.

#### 5.4 Tier 4 — Symbolic regression: the anti-black-box component

Symbolic regression (PySR) is run on field and synthetic data to *discover closed-form correlations specific to Baghewala*:

- a field-specific viscosity law  $\mu(T, \text{asphaltene content})$ ;
- a soak-time efficiency relation  $\eta(t_{\text{soak}}, V_{\text{steam}})$ ;
- a rod-failure hazard expression in terms of computed load range and compression duration.

The output of this tier is a **literal equation** — printable, dimensionally checkable, and capable of being signed off by a reservoir engineer and added to the field's operating manual. It is the cleanest possible contrast with a submission whose deliverable is a feature importance bar chart.

#### 5.5 Fault diagnosis by inverse simulation, not classification

The conventional approach is a convolutional network trained on dynamometer card images. That produces a label with no mechanism. The approach here produces a mechanism:

1. Convert the surface card to a downhole card using the Gibbs inverse solve.
2. Extract *physics features*: peak and minimum rod load, net stroke, pump fillage, card area (work per stroke), duration in compression, traveling and standing valve opening positions, load ratio.
3. Run a **parameter-fitting inverse solve** — find the ( $\mu$ , gas void fraction, leakage, tag depth) that best reproduces the observed card.

**The diagnosis is the fitted parameter set.** The output reads: “Fillage 61%; best-fit gas void fraction 0.34; pump intake pressure below bubble point — gas interference, not fluid pound.” That is a mechanism with numbers, not a class label. A gradient-boosted classifier with SHAP attribution is retained as a fast first-pass screen for real-time throughput, but the authoritative diagnosis always comes from the inverse solve.

## 6 The Optimization Layer

### 6.1 Real-time SRP control by Model Predictive Control

MPC is non-black-box by construction: it optimises an explicit objective, subject to explicit constraints, over an explicit model. Every recommendation can be decomposed into “this constraint became binding”.

**Decision variables.** SPM, stroke length, and the VFD velocity profile *within* the stroke (upstroke/downstroke asymmetry).

**Objective.** Maximise net fluid production, less electrical energy cost, less amortised failure risk computed from the Goodman fatigue accumulation.

| Hard constraint                               | Physical source                       |
|-----------------------------------------------|---------------------------------------|
| $\text{FMI}(z) > 0.15$ for all $z$            | rod float criterion, Section 4.5      |
| Peak and minimum load within modified Goodman | rod fatigue life                      |
| Pump fillage $> 85\%$                         | avoids fluid pound and impact loading |
| Gearbox torque $\leq$ rating                  | surface equipment limit               |
| Motor loading within VFD envelope             | electrical limit                      |
| Pump intake pressure $>$ NPSH requirement     | cavitation avoidance                  |

**The asymmetric stroke insight.** In heavy oil the *downstroke* is the constrained half of the cycle, not the upstroke — which is the reverse of conventional light-oil intuition. The correct response is therefore a slow, controlled downstroke that eliminates rod float and impact loading, paired with a faster upstroke that recovers cycle time and production rate.

A fixed-speed unit cannot do this. A VFD with an intra-stroke velocity profile can, and the FMI relation of Section 4.5 specifies exactly what profile to command as a function of  $\mu(T_{\text{pump}})$ . This single measure plausibly addresses the rod failure and pump unsettling objectives on its own, while *increasing* rather than sacrificing production.

### 6.2 CSS cycle design by Bayesian optimization over the physics model

$$\max_{V_s, P_{\text{inj}}, t_{\text{soak}}} \text{NPV} = \sum_t \frac{R_o q_o(t) - C_{\text{steam}} V_s - C_{\text{energy}} E(t) - C_{\text{fail}} \lambda(t)}{(1+r)^t} \quad (9)$$

subject to formation fracture pressure, boiler capacity, casing and cement thermal limits, and minimum cycle economics. Solved by Bayesian Optimization with a Gaussian Process surrogate seeded with a physics prior — convergence in roughly 50 simulator calls rather than thousands, and the posterior supplies an uncertainty map over the design space, so the system can report *how confident* a recommendation is rather than only what it is.

### 6.3 Production cut-off as an optimal stopping problem

This decision carries the largest economic leverage in the operation and deserves to be the intellectual centrepiece of the solution.

It has exactly the structure of a **renewal–reward process**. Let  $\pi(t)$  be the instantaneous net profit rate of the current cycle, and let  $\bar{\pi}^*$  be the maximum achievable *cycle-average* profit rate of a fresh optimized cycle — including injection cost and the non-producing soak period. The optimal policy is:

Stop producing and re-inject when  $\pi(t) \leq \bar{\pi}^*$

Continue only while the current well out-earns the average of starting over. Because  $\pi(t)$  falls

monotonically as  $\bar{T}(t)$  declines, the crossing point is unique and computable directly from the twin.

This replaces “we usually stop around day 60” with a defensible, recomputed threshold that responds daily to oil price, gas price, and the EnKF-updated reservoir parameters. It is simple enough to explain to a field operator in one sentence, and it is the most direct available lever on Steam–Oil Ratio.

#### 6.4 The safety envelope

A separate, rule-based layer sits between the optimizer and SCADA, enforcing only hard physics bounds. It can veto or clamp any setpoint, and it logs every intervention with the binding constraint named. The optimizer is never in unmediated control of equipment worth more than the software controlling it. Vetoes are also a training signal: a controller that is repeatedly clamped at the same constraint is telling you where the well’s real limit sits.

## 7 Data Strategy

The field has genuine operational history, but no team will have it in hand during the hackathon. Owning that gap with a concrete plan reads as competence; pretending otherwise does not.

1. **Synthetic twin generator.** Run the forward physics with parameters sampled from published Jodhpur Sandstone and Rajasthan heavy-oil literature. Inject realistic sensor noise, calibration drift, communication dropouts and seeded faults. This yields thousands of labelled CSS cycles and millions of dynamometer cards — enough to train and stress every component.
2. **Physics-first training order.** All models are trained on synthetic data generated by the same governing equations they are constrained by. They cannot be wildly wrong, because the physics bounds them before any real data arrives.
3. **Calibration path.** The tags to be bound on day one of a real deployment are specified precisely: OPC-UA from the VFD, Modbus from well-pad RTUs, steam mass flow totaliser, monthly well test results, and rod failure work orders. The EnKF absorbs each of these, and parameters migrate from prior to posterior with a documented collapse in uncertainty.
4. **Validation protocol.**
  - leave-one-cycle-out history matching on CSS records;
  - an independent energy-balance closure check (injected enthalpy versus accounted-for enthalpy) as a hard model audit;
  - dynamometer card reconstruction error against held-out cards;
  - a **counterfactual backtest** — run the optimizer over historical cycles and report what it would have recommended versus what was actually done, and the implied difference in SOR, energy per barrel and rod failure rate.

## 8 Technology Stack

| Layer                      | Choice                                                                    |
|----------------------------|---------------------------------------------------------------------------|
| <b>Physics solvers</b>     | Python, NumPy / SciPy; Numba for the wave-equation finite-difference loop |
| <b>PINN and autodiff</b>   | JAX (PyTorch acceptable alternative)                                      |
| <b>Data assimilation</b>   | FilterPy, with a custom Ensemble Kalman Filter implementation             |
| <b>Symbolic regression</b> | PySR                                                                      |
| <b>Surrogate modelling</b> | GPyTorch and BoTorch for Bayesian Optimization                            |
| <b>Predictive control</b>  | CasADi with IPOPT                                                         |
| <b>Time-series storage</b> | TimescaleDB (InfluxDB acceptable alternative)                             |
| <b>Streaming ingest</b>    | MQTT / OPC-UA gateway feeding Kafka                                       |
| <b>Backend API</b>         | FastAPI                                                                   |
| <b>Frontend</b>            | React with Plotly; 3D wellbore visualisation in Three.js                  |
| <b>Edge deployment</b>     | containerised twin on an industrial PC at the well pad                    |

**Edge capability is not optional for Rajasthan.** Connectivity at remote well pads is unreliable. A controller that stops controlling when the network link drops is not a controller. The full physics core, the FMI monitor and the safety envelope must run locally and autonomously; the cloud handles fleet

analytics, cross-well learning and long-horizon CSS planning.

## 9 What the Operator Actually Sees

Every recommendation carries four mandatory elements. This is the explainability contract, and it is enforced at the interface level — a recommendation that cannot populate all four fields is not displayed.

### RECOMMENDATION • Well BGW-07 • 14:32

**Reduce SPM 6.2 → 4.8; downstroke velocity –35%**

|                           |                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------|
| <b>Why</b>                | Float Margin Index at 620 m has fallen to 0.08, below the 0.15 operating limit.                |
| <b>Driver</b>             | Pump-intake temperature 58 °C → 51 °C over the last 96 hours; $\mu$ risen 340 → 890 cP.        |
| <b>Governing relation</b> | $(S \cdot N)_{\max} \propto 1/\mu$ — annular viscous drag versus buoyed rod weight.            |
| <b>Expected effect</b>    | –7% gross fluid; –31% peak compression load; rod fatigue life $\times 2.1$ ; net NPV positive. |
| <b>Confidence</b>         | 87%. EnKF posterior on $\mu$ : $\pm 120$ cP.                                                   |
| <b>Cycle status</b>       | Day 41 of production. Cut-off threshold projected at day $57 \pm 4$ .                          |

No SHAP plot appears anywhere in that card. An engineer can check the arithmetic, disagree with a specific number, and tell you which one — which is exactly the interaction that gets a system trusted and kept switched on.

## 10 Traceability: Requirements to Components

| Required outcome                                              | Delivering component                                                                                               |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Optimize CSS cycle parameters                                 | Marx–Langenheim / Boberg–Lantz core (4.1) driving NPV Bayesian optimization (6.2)                                  |
| Predict reservoir heating, cooling and production performance | Thermal decline model (4.1) with PINN surrogate (5.3), corrected by EnKF assimilation (5.1)                        |
| Continuously optimize SRP stroke speed and SPM                | Model Predictive Control (6.1) on the Gibbs rod model, driven by $\mu(T_{\text{pump}})$                            |
| Detect rod floating; minimise impact loading                  | Float Margin Index (4.5) as a live monitored scalar and a hard MPC constraint; asymmetric VFD stroke profile (6.1) |
| Improve pump efficiency and equipment reliability             | Inverse card diagnosis (5.5); fillage constraint; Goodman fatigue accounting (4.5)                                 |
| Optimize steam and energy consumption; reduce cost            | Optimal-stopping cut-off rule (6.3); NPV objective including steam and electrical cost                             |
| Reduce Steam–Oil Ratio                                        | Cut-off rule (6.3) plus soak-time and steam-volume optimization (6.2)                                              |
| Lower energy per barrel                                       | MPC energy term (6.1); elimination of wasted work against float and fluid pound                                    |
| Reduce rod failures and pump unsetting                        | FMI constraint, Lubinski buckling check, Goodman envelope (4.5); slow controlled downstroke (6.1)                  |
| Data-driven and predictive decision making                    | Full assimilation loop (5.1) with uncertainty-quantified recommendations and counterfactual backtesting (7)        |

## 11 Build Roadmap

| Phase | Deliverable                                                                                                                                              |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Physics engine: Marx–Langenheim, Boberg–Lantz, Ramey, and the Gibbs wave equation, each validated against published textbook cases with documented error |
| 2     | Synthetic data generator and the live FMI monitor; demonstrate rod float onset prediction ahead of occurrence                                            |
| 3     | EnKF assimilation loop; show estimated parameters converging on held-out synthetic ground truth with collapsing uncertainty bands                        |
| 4     | MPC controller and safety envelope; closed-loop demonstration with viscosity ramped through a full cooling cycle                                         |
| 5     | CSS optimizer and optimal-stopping cut-off rule; NPV and SOR comparison against historical practice                                                      |
| 6     | Dashboard, 3D wellbore visualisation, counterfactual backtest, and the explainability card interface                                                     |

**Phases 1 and 2 alone already constitute a stronger demonstration than most complete submissions,** because rod float is the failure mode that actually costs Oil India money at Baghewala, and a computed onset criterion for it is something very few teams will bring. Build depth-first in that order rather than breadth-first across a dashboard.

## 12 Why This Approach Wins

- 1. It works in the sparse-data regime that actually exists.** Thirty to eighty CSS cycles will overfit a deep model. Parameter estimation over a constrained physics model does not have that failure mode.
- 2. It extrapolates, because CSS optimization requires extrapolation.** Asked about a steam volume never previously attempted, a statistical model guesses and a conservation equation computes. The problem statement explicitly asks to move beyond historical practice — which rules out a purely historical model.
- 3. It is auditable end to end.** Every setpoint traces to a named equation and an identifiable binding constraint. That is the precondition for an operator actually leaving the system in automatic mode rather than overriding it.
- 4. It survives the domain expert's interrogation.** The evaluators are petroleum engineers, not ML researchers. Depth on the oilfield reasoning — what damping coefficient, what heat loss model, what fatigue envelope — will land considerably harder than an exotic model architecture.
- 5. The learning components are themselves engineering outputs.** A drifting skin estimate is an asphaltene deposition report. A discovered symbolic correlation is a field-specific equation that can enter the operating manual. The system produces knowledge about Baghewala, not just predictions for it.

*A digital twin earns its name by being wrong in measurable ways and correcting itself. The physics says what should happen; the field says what did; the difference, tracked as a named parameter with an uncertainty band, is where all the engineering value lives.*